

CSE 4104 – Software Development III

Initial Project Information

Field	Information
Official Team Name	CSE4104-7C-T02
Section	7C
Project Title (Initial Idea)	AI Prompt Engineering Learning Hub
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Full Member List with IDs

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Preliminary Project Description

The AI Prompt Engineering Learning Hub is a web-based educational platform designed to help users learn and improve prompt engineering skills for modern AI systems such as ChatGPT and Gemini. The platform will provide interactive prompt testing, AI-generated feedback, prompt optimization suggestions, and learning resources for students and beginners interested in prompt engineering.

Why AI is Needed in the Project

Artificial Intelligence is the core component of this system because the platform depends on AI models to analyze prompts, evaluate prompt quality, generate optimization suggestions, and provide intelligent feedback in real time. Without AI integration, the platform would only function as a static learning website instead of an interactive prompt engineering environment.

Initial Technology Stack

System Component	Technology
Frontend	React.js + Tailwind CSS
Backend	Node.js + Express.js
Database	MongoDB / PostgreSQL
AI Tools	Google Gemini API / OpenAI API

Similar Existing Systems / Research

The team researched several AI platforms and prompt learning resources including ChatGPT, Google Gemini, Claude AI, PromptHero, and Learn Prompting. These systems provide AI interaction or prompt examples, but most do not provide a complete interactive learning hub focused on prompt engineering education, optimization, and guided experimentation.